Romanian palatalization without tears

In Romanian, coronal and dorsal obstruents become palatalized when followed by certain suffixes beginning with a front vowel, such as noun plurals in *-i* (NB: /i/ ⇝ [ʲ] word-finally).

(1)

rus	ruʃʲ	‘Russian(s)’	brad	brazʲ	‘fir(s)’	munte	muntsʲ	‘mountain(s)’
prost	proʃtʲ	‘fool(s)’	milog	milodʒʲ	‘beggar(s)’	pisikə	pisitʃʲ	‘cat(s)’

These *palatalization* mutations are straightforwardly handled by a series of ordinary phonological rules or constraints. However, additional observations have led some (e.g., Chitoran 2001) to posit that palatalization is morphophonological in part. Palatalization exhibits *non-derived environment blocking* (NDEB): it does not apply when the obstruent target and front vowel trigger are underlyingly adjacent and tautomorphemic; e.g., [kilogram] ‘kilogram’, [gimpe] ‘thorn’. But there are also some derived exceptions, such as some agent nominals in *-ist*; e.g., [fok-ist] ‘stoker’, [taŋk-ist] ‘tank driver’, [drog-ist] ‘druggist’, [jog-ist] ‘yogi’ cf. the palatalization observed in [kajak, kajatʃ-ist] ‘kayak(er)’, [stɨŋg, stɨŋdʒ-ist] ‘left(ist)’. Steriade (2008) further claims that nouns that select plural allomorphs which trigger palatalization (e.g., *-i*) tend to select palatalizing allomorphs of derivational suffixes—like the homophonous *-i* verbalizer as in (2a)—and nouns which select non-palatalizing plural allomorphs (e.g., *-uri*) tend to select non-palatalizing verbalizer allomorphs (e.g., *-a* or *-ui*) as in (2b).

(2)

a.	kolak	kolatʃʲ	‘kalach(es)’	ɨŋkolatʃʲ	‘to coil’
	verde	verzʲ	‘green(s)’	ɨnverzʲ	‘to become green’
b.	fok	fokurʲ	‘fire(s)’	ɨnfoka	‘to fire up’
	tɨrg	tɨrgurʲ	‘market(s)’	tɨrgui	‘to go shopping’

We first consider the productivity of the different forms of palatalization. Productivity analyses are conducted using a lexicon extracted from Wiktionary and the DEX dictionary (Academia Română 1998) filtered by frequency (Goldhahn et al. 2012) to simulate realistic lexical knowledge (e.g., Kodner 2019). Below we give the type frequencies of noun plural palatalization (*n*: # of stems eligible for palatalization, *e*: # of stems which are exceptions to palatalization) for the 1,000 most common nouns in our sample, with majority-rules (Nida 1949: 14) and Tolerance Principle (Yang 2005, 2016) calculations, showing all coronal- and dorsal-obstruent-final stems when followed by heterosyllabic *i* or *e*. Under both thresholds, there is productive palatalization of *c* and *g* before *i* and *e*, and of *t*, *d*, and *s* before *i*.

(3)

	n	*e*	majority?	tolerated?		*n*	*e*	majority?	tolerated?
…c-i…	28	0	✓	✓	*…c-e…*	14	0	✓	✓
…g-i…	17	0	✓	✓	*…g-e…*	3	0	✓	✓
…t-i…	207	0	✓	✓	*…t-e…*	229	229	✗	✗
…d-i…	23	0	✓	✓	*…d-e…*	36	36	✗	✗
…s-i…	11	0	✓	✓	*…s-e…*	44	44	✗	✗
…z-i…	2	1	(n.a.)	(n.a.)	*…z-e…*	48	48	✗	✗
…st-i…	4	0	✓	✓	*…st-e…*	0	0	(n.a.)	(n.a.)
…ct-i…	3	0	✓	✓	*…ct-e…*	22	22	✗	✗

Rasin (2025) provides an analysis of Romanian NDEB using a form of *inalterability as prespecification* (e.g., Inkelas and Cho 1993; Kiparsky 1993; Buckley 1994; Inkelas 1995; Inkelas et al. 1997; Inkelas 2000, 2014; Gorman 2026). In this pattern, segments which undergo palatalization are underspecified and thus mutable w.r.t. feature-filling palatalization rules, but prespecified segments are inalterable w.r.t. these rules. We focus on nouns ending in *c* or *g* for the sake of brevity, though this account generalizes to coronals as

well. To implement the desired feature-filling behavior, we use unification rules (Bale et al. 2014). Rules (4–5) map underspecified /K, G/ to [tʃ, dʒ] (resp.) before front vowels and to [k, g] elsewhere. While these rules intensionally target many segments, unification is defined so that it does not overwrite existing feature specifications, so prespecified segments like /k, g/ are inalterable w.r.t. both rules.

(4) Dorsal palatalization:

$$\begin{bmatrix} -\textsc{Syllabic} \\ +\textsc{Consonantal} \\ -\textsc{Sonorant} \\ -\textsc{Approximant} \\ -\textsc{Nasal} \\ \dots \end{bmatrix} \sqcup \begin{Bmatrix} +\textsc{Coronal} \\ -\textsc{Dorsal} \\ -\textsc{Anterior} \\ +\textsc{Distributed} \\ +\textsc{Strident} \end{Bmatrix} __ / \begin{bmatrix} +\textsc{Syllabic} \\ +\textsc{Front} \end{bmatrix}$$

(5) Dorsal default fill-in:

$$\begin{bmatrix} -\textsc{Syllabic} \\ +\textsc{Consonantal} \\ -\textsc{Sonorant} \\ -\textsc{Approximant} \\ -\textsc{Nasal} \\ \dots \end{bmatrix} \sqcup \begin{Bmatrix} -\textsc{Coronal} \\ +\textsc{Dorsal} \\ -\textsc{Continuant} \end{Bmatrix}$$

This underspecification analysis can also be used to account for the suffix allomorphy patterns described by Steriade. Assuming inside-out spell-out (e.g., Bobaljik 2000), we propose that spell-out of the noun plural suffix Num[Pl] and verbalizer *v* is conditioned on the underlying phonological form of the root. Roots prespecified for −Coronal, +Dorsal, and −Continuant select the non-palatalizing allomorphs *-uri* and *-a*. On the other hand, palatalizing roots lack these specifications and thus fall through to the elsewhere allomorph *-i*. Similar patterns govern the spell-out of the *v* allomorphs.

(6) Num[Pl] $\Longleftrightarrow$ $\begin{cases} \text{/-uri/} & \begin{bmatrix} -\textsc{Coronal} \\ +\textsc{Dorsal} \\ -\textsc{Continuant} \end{bmatrix} __ \\ \text{/-i/} & \text{elsewhere} \end{cases}$

(7) *v* $\Longleftrightarrow$ $\begin{cases} \text{/-a/} & \begin{bmatrix} -\textsc{Coronal} \\ +\textsc{Dorsal} \\ -\textsc{Continuant} \end{bmatrix} __ \\ \text{/-i/} & \text{elsewhere} \end{cases}$

The interaction between allomorph selection and palatalization is illustrated in the derivations below, which begin immediately after root spell-out.

(8)

UR:	fok	fok⌢Num[Pl]	koloK	koloK⌢Num[Pl]	miloG	miloG⌢*v*
Rule (6):		fokuri		koloKi		
Rule (7):						miloGi
Rule (4):				kolatʃi		milodʒi
Rule (5):			kolak		milog	
Desyllabification:		fokurʲ		kolatʃʲ		milodʒʲ
SR:	fok	fokurʲ	kolak	kolatʃʲ	milog	milodʒʲ

We tentatively propose that noun plural and verbalizer selection could appear to align simply because both are conditioned by a similar phonological context. This shows Steriade's *inflection dependence* pattern can be generated without the explicit paradigmatic dependencies her analysis presupposes. However there are some empirical doubts about the proposed connection between the two allomorphy rules: Fisher's exact tests find no significant correlation between roots' allomorphs of Num[Pl] and *v*, and this result holds when considering dorsals and coronals separately or grouped together.

(9)

	c, g				*t, d, s*				*c, g* + *t, d, s*			
	-i	*-a*	OR	*p*	*-i*	*-a*	OR	*p*	*-i*	*-a*	OR	*p*
-i, -e	21	16	1.49	.553	26	73	0.92	.846	47	89	1.11	.874
-uri	7	8			14	36			21	44		

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